example, there are technical challenges to define a notion of probability in a context that involves an infinite number of copies of each of the observers whose measurements are supposed to be subject to those probabilities. If a given observer is really one of many copies, in what sense can we say that he or she has a particular probability to measure this or that outcome? Who really is “he” or “she”? Each copy of the observer will measure—with probability 1—whatever outcome is slated for the particular copy of the universe in which he or she resides, so the whole probabilistic framework requires (and has been given, and continues to be given) careful scrutiny in the Many Worlds framework. Moreover, on a more technical note, the mathematically inclined reader will realize that, depending on how one precisely defines the Many Worlds, a preferred eigenbasis may need to be selected. But how should that eigenbasis be chosen? There has been a great deal of discussion and much written on all these questions, but to date there are no universally accepted resolutions. The approach based on decoherence, discussed shortly, has shed much light on these issues, and has offered particular insight into the issue of eigenbasis selection.

13. The Bohm or de Broglie–Bohm approach has never received wide attention. Perhaps one reason for this, as pointed out by John Bell in his article “The Impossible Pilot Wave,” collected in *Speakable and Unspeakable in Quantum Mechanics*, is that neither de Broglie nor Bohm was particularly fond of what he himself had developed. But, again as Bell points out, the de Broglie–Bohm approach does away with much of the vagueness and subjectivity of the more standard approach. If for no other reason, even if the approach is wrong, it is worth knowing that particles can have definite positions and definite velocities at all times (ones beyond our ability, even in principle, to measure), and still conform fully to the predictions of standard quantum mechanics—uncertainty and all. Another argument against Bohm’s approach is that the nonlocality in this framework is more “severe” than that of standard quantum mechanics. By this it is meant that Bohm’s approach has nonlocal interactions (between the wavefunction and particles) as a central element of the theory from the outset, while in quantum mechanics the nonlocality is more deeply buried and arises only through nonlocal correlations between widely separated measurements. But, as supporters of this approach have argued, because something is hidden